

Evanns Morales-Cuadrado

Robotics PhD Candidate | Fulbright Specialist | Georgia Institute of Technology

U.S. Citizen

CONTACT

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SUMMARY

Robotics PhD candidate in Dr. Samuel Coogan’s FACTS Lab at Georgia Tech and graduate research intern at Sandia National Laboratories, developing safe, computationally efficient planning and control for aerial and multi-robot systems. At Sandia, built a kinetic-intercept benchmark suite for the U.S. gov-ernment, approved for Unclassified Unlimited Release and submitted to ICRA 2027, and designed a new intercept control law benchmarked against guidance-law, NMPC, and reinforcement-learning methods. Leads a collaboration with Brazilian partners at Lactec that produced an ICRA 2027 submission and is bringing \$20K to Georgia Tech.

- TECHNICAL SKILLS
- **Programming Languages:** Python, C/C++, C#/.NET, Matlab/Simulink
 - **Robotics & Control Frameworks:** ROS 2, NATS, PX4, ArduPilot, acados, CasADi
 - **Machine Learning & Simulation:** JAX, PyTorch, MuJoCo, Reinforcement Learning
 - **Development & DevOps Tools:** Docker, Continuous Integration, Git/GitHub
 - **Hardware & Experimentation:** UAV assembly, repair, and deployment
 - **Motion-Capture:** OptiTrack and Vicon
 - **Languages:** Spanish (Native), English (Native), Portuguese (B1)

EDUCATION

Georgia Institute of Technology2022-Present

Ph.D. in Robotics

Doctoral Minor in Optimization Methods

- GPA: 4.0/4.0
- Advisor: Dr. Samuel Coogan
- Thesis Proposal Approved (ABD)
- Goizueta Fellow

University of Texas at Arlington2018-2022

Honors Bachelor of Science in Electrical Engineering

Minor in Mathematics

Minor in Physics

Certificate in Unmanned Vehicle Systems

- GPA: 3.9/4.0 Summa Cum Laude
- Full National Merit Scholarship Awarded
- Electrical Engineering Honors Scholar

WORK EXPERIENCE

Year-Round Graduate Research Intern | AI for AutonomyMay 2026 - Present

Sandia National LaboratoriesAlbuquerque, NM

- Applied reinforcement learning to autonomy problems in support of national security
- Designed and deployed robotics and automation infrastructure that enables multi-robot experiments
- Built C#/.NET networking frameworks and execution scripts that coordinate communication and task execution across multi-robot systems, now used team-wide

Graduate Research Assistant | FACTS Lab

August 2022 - Present

Georgia Institute of Technology

Atlanta, GA

- Advised by Dr. Samuel Coogan
- Developed a lightweight tracking control law that beat PX4's native tracker in aggressive quadrotor flight and matched NMPC's tracking accuracy at far lower computation time and CPU energy
- Developed safe planning and control methods that combine reachability analysis, formal methods, optimization, and learning to generate fast, provably safe trajectories under unknown disturbances
- Built the lab's real-time flight-test stack, enabling repeatable hardware validation

Teaching Assistant | Vertically Integrated Project in Robotics

January 2024 - May 2026

Georgia Institute of Technology

Atlanta, GA

- Mentored about 20 undergraduates per semester on hands-on robotics research, with multiple mentees going on to robotics PhD programs
- Designed and delivered undergraduate lectures on ROS, Git/GitHub, Linux, and control theory, preparing students for research on hardware

Research Assistant | Autonomous Systems Lab

November 2021 - May 2022

University of Texas at Arlington

Arlington, TX

- Advised by Dr. Frank Lewis
- Contributed to reinforcement-learning quadrotor control research under Dr. Frank Lewis, then ranked 12th in the USA and 23rd worldwide
- Completed UTA Honors College thesis on lightweight Newton-Raphson tracking control for computationally constrained aerial systems

Teaching Assistant | Graduate-Level Intelligent Systems Course

January 2022 - May 2022

University of Texas at Arlington

Arlington, TX

- Selected as an undergraduate to TA a graduate-level course
- Led exam review sessions and graded homework, exams, and projects

Research Assistant | Dynamical Networks and Control Lab

October 2019 - May 2022

University of Texas at Arlington

Arlington, TX

- Advised by Dr. Yan Wan
 - Conducted original research on learning-based minimum-time and minimum-energy multi-vehicle path planning using ROS and OpenCV
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SELECTED PUBLICATIONS	E. Morales-Cuadrado , S. Coogan, Z. Kakish, and A. Parikh. Submitted to <i>2027 IEEE International Conference on Robotics and Automation (ICRA)</i> . (title redacted until the double-blind review process is complete)	
	L. Pinto*, E. Morales-Cuadrado* , S. Coogan, M. A. Wehrmeister, L. Caselatto, and C. E. Magrin. Submitted to <i>2027 IEEE International Conference on Robotics and Automation (ICRA)</i> . *Equal contribution. (title redacted until the double-blind review process is complete)	
	M. Cao, A. Harapanahalli, E. Morales-Cuadrado , and S. Coogan, “Trajectory Tracking for Systems with Unknown Time-Varying Disturbances.” <i>IEEE Open Journal of Control Systems (OJ-CSYS)</i> , 2026.	
	E. Morales-Cuadrado , “Computationally Efficient and Safe Control for Aerial Robotic Systems under Threat and Disturbance.” Ph.D. Thesis Proposal, Georgia Institute of Technology, 2026. pdf available . slides available .	
	E. Morales-Cuadrado , L. Chung, S. Kousik, and S. Coogan, “RTD–RAX: Fast, Safe Trajectory Planning for Systems under Unknown Disturbances.” Joint Submission to 2026 Modeling, Estimation and Control Conference (MECC) and ASME Letters in Dynamic Systems and Control (ALDSC). (accepted to both) preprint available .	
	E. Morales-Cuadrado , L. Baird, Y. Wardi, and S. Coogan, “Lightweight Tracking Control for Computationally Constrained Aerial Systems with the Newton-Raphson Method.” <i>IEEE Transactions on Control Systems Technology (TCST)</i> , 2026. preprint available .	
RESEARCH INTERESTS	L. Baird, E. Morales-Cuadrado , and S. Coogan, “Runtime Assurance for Uncertain Systems from Interval Signal Temporal Logic.” Submitted to IFAC Nonlinear Analysis: Hybrid Systems (NAHS), 2025. (under revision) preprint available .	
	E. Morales-Cuadrado , C. Llanes, Y. Wardi, and S. Coogan, “Newton-Raphson Flow for Aggressive Quadrotor Tracking Control.” <i>2024 American Control Conference (ACC)</i> . preprint available .	
	• Safe Autonomy • Hardware Deployment • Unmanned Ground and Aerial Vehicles	• Advanced Nonlinear Control • Learning-Based Control • Trajectory Generation and Planning
	OPEN-SOURCE CONTRIBUTIONS	
	• PX4 Autopilot : Contributed documentation improvements to the official PX4 Autopilot manual. • Vicon4PX4 : Developed and open-sourced a ROS 2 C++ Vicon–PX4 interface enabling external vision fusion for indoor quadrotor flight experiments; adopted and showcased by Georgia Tech’s Indoor Flight Lab as part of its official codebase. Also created a parallel package for OptiTrack systems.	
	SERVICE & LEADERSHIP	
Co-Founder and President of Puerto Rican Student Association		August 2024 - Present
Georgia Institute of Technology		Atlanta, GA
• Addressed the need for an organization to help Puerto Rican students feel at home and stay connected to our culture • Formed a three-member co-founder board and identified a faculty advisor • Raised funds for the organization and recruited members from campus • Hosted professional, social, and cultural events to meet the needs of the growing Puerto Rican student population at Georgia Tech		
Undergraduate Research Mentor		August 2021 - May 2022
University of Texas at Arlington		Arlington, TX
• As the lead undergraduate student in Dr. Frank Lewis’ Autonomous Systems Lab, mentored junior undergrads in controls research and guided them on a research project in ground vehicle control • Guided students in Robot Operating System, Python, Linux, and Control Systems		
Teaching Assistant Graduate-Level Intelligent Systems Course		January 2022 - May 2022
University of Texas at Arlington		Arlington, TX
• Selected as an undergraduate to TA a graduate-level course • Led exam review sessions and graded homework, exams, and projects		

Eta Kappa Nu Public Relations Chair & Circuits I Tutor

University of Texas at Arlington

August 2020 - May 2022

Arlington, TX

- Executive board member of UT Arlington's IEEE-HKN chapter
- In charge of social media advertising for the honor society's events and outreach initiatives for the electrical engineering student body
- Volunteered to provide free tutoring as a service to electrical engineering students in Circuits I

**HONORS AND
AWARDS****Fulbright Specialist**

August 2026

Peer-review-selected to the U.S. Department of State's Fulbright Specialist Roster for a three-year term

Top-3 Finalist at a Deep Learning Research Symposium

November 2023

Professor-sponsored award for novel research in deep learning and robotics

Goizueta Fellowship at Georgia Tech

August 2022

Highly selective fellowship for graduate students of Latin-American descent

Summa Cum Laude Honors at University of Texas at Arlington

May 2022

Exceptional academic distinction in top GPA tier for graduation honors

Honors College Degree Requirements at University of Texas at Arlington

December 2021

Completed research projects in disparate fields throughout my undergraduate career with various professors and defended a senior capstone research project before the Honors College committee

Electrical Engineering Honors Scholar at University of Texas at Arlington

December 2021

Inaugural member of the electrical engineering honors cohort

IEEE-HKN International Electrical Engineering Honor Society Membership

August 2021

Selective undergraduate society recognizing outstanding academic achievement and community service

Research Experience for Undergraduates Fellowship Recipient

August 2020

Fellowship to sponsor promising young undergraduate researchers

Chance Vought Engineering and Science Endowment Scholarship

August 2020

Highly selective award given annually to three recipients across the entire College of Engineering

Toray Composite Materials America Inc. Sons and Daughters Scholarship

May 2019

Academic scholarship awarded to children of employees of Toray America

James E. Casey Scholarship Recipient

January 2019

Academic scholarship awarded to children of employees of UPS

James R. Hoffa Memorial Scholarship Recipient

May 2018

Academic scholarship awarded to children of Teamsters Union Members

Grapevine, TX Rotary Club Scholarship Recipient

May 2018

Local rotary club scholarship for academic and civically-minded scholars

National Hispanic Scholar

March 2018

Recognition of top national performers of Latin descent on the SAT

National Merit Scholar

March 2018

Recognition of top national performers on the SAT